

Problem Set 11
Due: May. 17, 2021

1. Section 4.5 Problem 41.

At a certain time, the number of people that enter an elevator is a Poisson random variable with parameter λ . The weight of each person is independent of every other person's weight, and is uniformly distributed between 100 and 200 lbs. Let X_i be the fraction of 100 by which the i th person exceeds 100 lbs, e.g., if the 7th person weighs 175 lbs., then $X_7 = 0.75$. Let Y be the sum of the X_i .

- (a) Find the transform associated with Y .
- (b) Use the transform to compute the expected value of Y .
- (c) Verify your answer to part (b) by using the law of iterated expectations.

2. Section 4.5 Problem 43.

A motorist goes through 4 lights, each of which is found to be red with probability $1/2$. The waiting times at each light are modeled as independent normal random variables with mean 1 minute and standard deviation $1/2$ minute. Let X be the total waiting time at the red lights.

- (a) Use the total probability theorem to find the PDF and the transform associated with X , and the probability that X exceeds 4 minutes. Is X normal?
- (b) Find the transform associated with X by viewing X as a sum of a random number of random variables.

3. Section 4.5 Problem 44.

Consider the calculation of the mean and variance of a sum

$$Y = X_1 + \cdots + X_N$$

where N is itself a sum of integer-valued random variables, i.e.,

$$N = K_1 + \cdots + K_M$$

Here $N, M, K_1, K_2, \dots, X_1, X_2, \dots$ are independent random variables, $N, M, K_1, K_2, \dots$ are integer-valued and nonnegative, $K_1, K_2, \dots$ are identically distributed with common mean and variance denoted $\mathbf{E}[K]$ and $\text{var}(K)$, and $X_1, X_2, \dots$ are identically distributed with common mean and variance denoted $\mathbf{E}[X]$ and $\text{var}(X)$.

- (a) Derive formulas for $\mathbf{E}[N]$ and $\text{var}(N)$ in terms of $\mathbf{E}[M]$, $\text{var}(M)$, $\mathbf{E}[K]$, $\text{var}(K)$.
- (b) Derive formulas for $\mathbf{E}[Y]$ and $\text{var}(Y)$ in terms of $\mathbf{E}[M]$, $\text{var}(M)$, $\mathbf{E}[K]$, $\text{var}(K)$, $\mathbf{E}[X]$, $\text{var}(X)$.

- (c) A crate contains M cartons, where M is geometrically distributed with parameter p . The i th carton contains K_i widgets, where K_i is Poisson-distributed with parameter μ . The weight of each widget is exponentially distributed with parameter λ . All these random variables are independent. Find the expected value and variance of the total weight of a crate.
4. (a) Let X, Y, Z be independent and uniformly distributed on $[0, 1]$. Find the density function of $X + Y + Z$.
- (b) If $\{X_r : r \geq 1\}$ are independent and uniformly distributed on $[0, 1]$, show that the density of $\sum_{r=1}^n X_r$ at any point $x \in (0, n)$ is a polynomial in x of degree $n - 1$.